Round 2: orbit-union τ-slots for the split and hybrid PDP, measured in ecbench

Study `research/pdp_split_slots_20261012` · branch `claude/pdp-split-dreg` (PR #1709) · 2026-10-12/19 · **stage diagnostic plus end-to-end ecbench measurement; no speedup claim.** Protocol `PROTOCOL.md` (registered before any measured run) with amendments 1–3 (three container reclaims; inputs unchanged). Analysis `ANALYSIS.json` (native `pdp_split_analyze`). Full tables: `README.md`. Run reports: `reports/run-*`.

Question

Round 1 decomposed over the standard subspace `V = span(1, z, …, z^(ℓ−1))` and projected costs. Round 2 runs the same split and hybrid systems as a real `ic.pipeline` oracle (`pdp-split`) in ecbench, on `V` and on its Frobenius closure `∪ σ^j(V)` with τ-slot systems and one column per signed Frobenius orbit, against matched rho on the same public targets.

Correctness

Every ecbench run recorded verified `[k]G = Q` (Parts A and C); all 576 Part B runs match an exhaustive enumeration exactly. The 20 n = 19 (workload, arm) pairs recorded by both interrupted sessions are bit-identical. Replay audits: `audits/`.

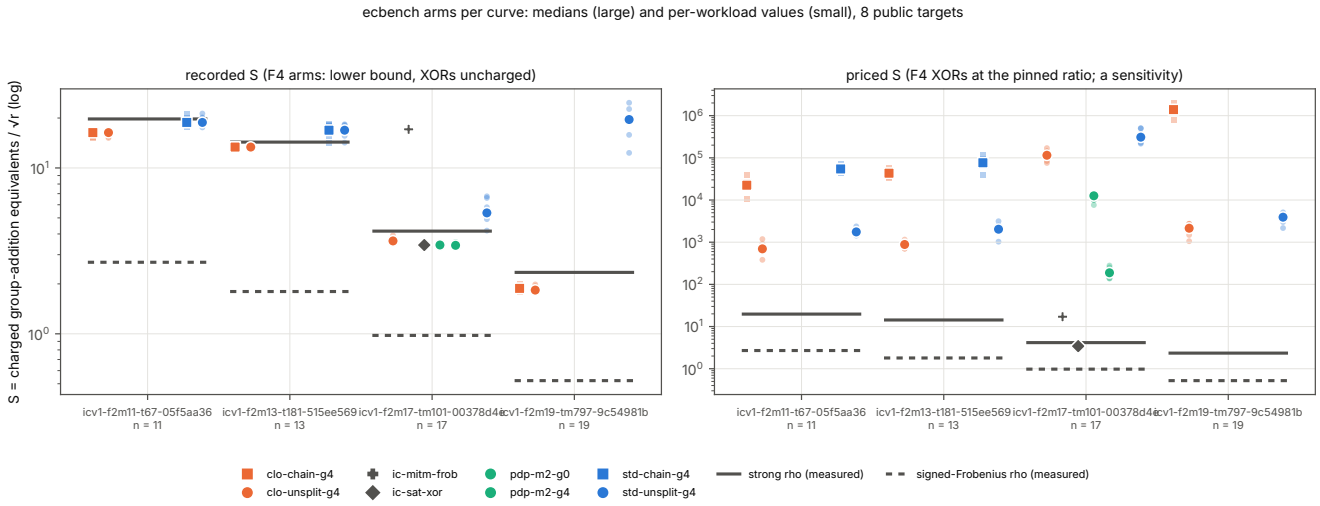


Figure 1: Recorded S (left; F4 and SAT arms are lower bounds, solver work uncharged) and priced S (right; F4 word XORs at the pinned ratio, a sensitivity) per arm and curve, against measured strong and signed-Frobenius rho. Medians large, per-workload values small. Sessions in `sessions/`; n = 19 censored as in amendment 3.

Results

Closure vs standard	On all four curves the closure solved 2.0–3.2× fewer systems per IC run and cost 0.35–0.54× in priced S (R1, R2 held), with 1.5–4.5× the yield per system and the same solving degree (R3, R4 held).
Against rho	Best F4 arm (closure, unsplit, g = 4): 36× strong rho at n = 11, 60× at 13, 889× at 19, 25,257× at 17 (ℓ = 5). No arm came close (R5 held).
Part C, n = 17	On the dimension-9 invariant base the fully charged pair table (`mitm-frobenius:m=2`) costs 4.18× strong rho; `pdp-split` m = 2 prices at 46× (g = 4) and 2,919× (g = 0); the SAT arm's lower bound is 0.86× (R7 held).
Accounting	Every F4/SAT arm's recorded S is at or below strong rho because solver work is uncharged: a lower bound, not a win.

Interpretation (labelled)

The Frobenius closure is the better base for split and hybrid systems: same per-system cost, more decompositions per system, by about the ordering factor predicted before the run. It moves the cost by a constant factor of 2–3 and does not change the comparison with rho; the fully charged pair table remains the cheapest complete IC on the curve where both ran. Limits: `r ≤ 2^18`, one round, L0 counts; n = 19 partly censored; XOR pricing omits F4's per-call work.